Property (4)

(i) $\sin^{-1}(x)+\cos^{-1}(x)=\frac{\pi}{2}$

(ii) $\tan^{-1}(x)+\cot^{-1}(x)=\frac{\pi}{2}$

(iii) $\sec^{-1}(x)+\operatorname{cosec}^{-1}(x)=\frac{\pi}{2}$

Property (5)

If $xy<1$, then

(i) $\tan^{-1}(x)+\tan^{-1}(y)=\tan^{-1}\left(\frac{x+y}{1-xy}\right)$

(ii) $\tan^{-1}(x)-\tan^{-1}(y)=\tan^{-1}\left(\frac{x-y}{1+xy}\right)$

Property (6)

$$\sin^{-1}(x)+\sin^{-1}(y)$$
$$=\sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)$$

Example 4.23

Find the principal value of

(i) $\sin^{-1}(1/2)$ (ii) $\tan^{-1}(-\sqrt{3})$

Solution

(i) Let $\sin^{-1}(1/2)=$ y where $-\frac{\pi}{2}\le y\le\frac{\pi}{2}$

$\therefore \sin y=(1/2)=\sin\left(\frac{\pi}{6}\right)$

$y=\frac{\pi}{6}$

The principal value of $\sin^{-1}(1/2)$ is $\frac{\pi}{6}$

(ii) Let $\tan^{-1}(-\sqrt{3})=y$ where $-\frac{\pi}{2}\le y\le\frac{\pi}{2}$

$\therefore \tan y=-\sqrt{3}=\tan\left(-\frac{\pi}{3}\right)$

$y=-\frac{\pi}{3}$

The principal value of $\tan^{-1}(-\sqrt{3})$ is $-\frac{\pi}{3}$

Example 4.24

Evaluate the following

(i) $\cos\left(\sin^{-1}\frac{5}{13}\right)$ (ii) $\tan\left(\cos^{-1}\frac{8}{17}\right)$

Solution

(i) Let $\left(\sin^{-1}\frac{5}{13}\right)=\theta$ ... (1)

$\sin\theta=\frac{5}{13}$

$\cos\theta=\sqrt{1-\sin^2\theta}$

$=\sqrt{1-\frac{25}{169}}$

$=\frac{12}{13}$ ... (2)

From (1) and (2), we get

Now $\cos\left(\sin^{-1}\frac{5}{13}\right)=\cos\theta=\frac{12}{13}$

(ii) Let $\left(\cos^{-1}\frac{8}{17}\right)=\theta$

$\cos\theta=\frac{8}{17}$

$\sin\theta=\sqrt{1-\cos^2\theta}$

$=\sqrt{1-\frac{64}{289}}$

$=\frac{15}{17}$ ... (2)

From (1) and (2), we get

Now $\tan\left(\cos^{-1}\frac{8}{17}\right)=\tan\theta=\frac{\sin\theta}{\cos\theta}$

$=\frac{\frac{15}{17}}{\frac{8}{17}}=\frac{15}{8}$

Example 4.25

Prove that

(i) $\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)=\tan^{-1}\left(\frac{2}{9}\right)$

(ii) $\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{3}{5}\right)=\tan^{-1}\left(\frac{27}{11}\right)$

Solution

(i) $\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)$

$=\tan^{-1}\left[\frac{\frac{1}{7}+\frac{1}{13}}{1-\left(\frac{1}{7}\right)\left(\frac{1}{13}\right)}\right]$

$=\tan^{-1}\left(\frac{20}{90}\right)$

$=\tan^{-1}\left(\frac{2}{9}\right)$

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